

Multiple Choice (10 marks)

1. What was the prevailing notion prior to the time of Lyell and Darwin?
 - (a) Earth is millions of years old, and populations are unchanging.
 - (b) Earth is millions of years old, and populations gradually change.
 - (c) Earth is a few thousand years old, and populations rapidly change.
 - (d) Earth is a few thousand years old, and populations are unchanging.
 - (e) Earth is billions of years old, and populations gradually change.
2. Which of the following plant cells undergoes programmed cell death to become functional?
 - (a) Epidermal cell
 - (b) Stomatal guard cell
 - (c) Root cap cell
 - (d) Xylem vessel member
 - (e) Phloem sieve tube member
3. Which of the following is the source of oxygen produced during photosynthesis?
 - (a) O_2
 - (b) CH_4 (Methane)
 - (c) CO_2
 - (d) H_2O
 - (e) N_2 (Nitrogen)
4. What features enable the ferns to grow to a much larger size than the mosses?
 - (a) Ferns grow larger because they can perform photosynthesis more efficiently than mosses
 - (b) Ferns grow larger due to higher nutrient content
 - (c) Ferns grow larger due to a stronger stem
 - (d) Ferns grow larger due to their ability to move to more fertile areas
 - (e) Mosses are limited in size by their shorter life cycle compared to ferns
5. The principle inorganic compound found in living things is
 - (a) oxygen
 - (b) calcium

- (c) carbon
- (d) potassium
- (e) phosphorus

True / False (10 marks)

Answer True or False

6. True or False: High relative humidity increases the rate of transpiration from plants by enhancing evaporative water loss from leaf surfaces.
7. True or False: A change in a neuron membrane potential from +50 millivolts to -70 millivolts is an action potential.
8. True or False: The process of molting in crustaceans is controlled by hormonal influences in response to environmental factors.
9. True or False: Epidermis in plants functions like animal skin by allowing for gas exchange and nutrient transport within the plant.
10. True or False: Schizophrenia is primarily caused by environmental factors, with little to no genetic influence.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how inbreeding and loss of genetic variation contribute to the vulnerability of small populations and their potential descent into an extinction vortex.
12. Discuss the differences in energy release between combustion and aerobic respiration, focusing on the forms of energy produced and their implications for biological systems.

End of Paper